

ESM3. Classification Reports

Tables I to VIII present partial classification reports for a selection of configurations, including the best-performing model (DenseNet201 + Fine-tuned + Attention) and the best baseline model (ResNet50 + Frozen + No Attention). We have selected specific rows to emphasize key results, with ellipses (...) indicating that additional rows exist before and after the displayed subset.

TABLE I: Classification Report for DenseNet201 + Fine-tuned + No Attention

Class Name	Value (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	1.0000 (0.0000)	0.9949 (0.0089)	0.9974 (0.0045)	65.0000 (0.0000)
Angele Ellis	0.9744 (0.0444)	1.0000 (0.0000)	0.9867 (0.0231)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.6667 (0.5774)	0.6667 (0.5774)	0.6667 (0.5774)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	3.0000 (0.0000)
Youssef Bey Al-Helou	1.0000 (0.0000)	0.8333 (0.2887)	0.8889 (0.1925)	2.0000 (0.0000)
Father Elias	0.6667 (0.5774)	0.6667 (0.5774)	0.6667 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9862 (0.0035)	0.9862 (0.0035)	0.9862 (0.0035)	0.9862 (0.0035)
macro avg	0.9638 (0.0205)	0.9561 (0.0215)	0.9557 (0.0226)	2849.0000 (0.0000)
weighted avg	0.9863 (0.0041)	0.9862 (0.0035)	0.9852 (0.0041)	2849.0000 (0.0000)

TABLE II: Classification Report for the Best Model (DenseNet201 + Fine-tuned + Attention)

Class Name	Value (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	1.0000 (0.0000)	0.9949 (0.0089)	0.9974 (0.0045)	65.0000 (0.0000)
Angele Ellis	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	0.9167 (0.1443)	1.0000 (0.0000)	0.9524 (0.0825)	3.0000 (0.0000)
Youssef Bey Al-Helou	1.0000 (0.0000)	0.8333 (0.2887)	0.8889 (0.1925)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9905 (0.0009)	0.9905 (0.0009)	0.9905 (0.0009)	0.9905 (0.0009)
macro avg	0.9819 (0.0039)	0.9730 (0.0040)	0.9744 (0.0028)	2849.0000 (0.0000)
weighted avg	0.9912 (0.0006)	0.9905 (0.0009)	0.9902 (0.0007)	2849.0000 (0.0000)

TABLE III: Classification Report for the Best Baseline Model (ResNet50 + Frozen + No Attention)

Class Name	Value (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.8490 (0.0410)	0.9128 (0.0622)	0.8787 (0.0355)	65.0000 (0.0000)
Angele Ellis	0.5541 (0.0495)	0.8889 (0.1273)	0.6799 (0.0565)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.1667 (0.2887)	0.1667 (0.2887)	0.1667 (0.2887)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	0.3333 (0.5774)	0.1111 (0.1925)	0.1667 (0.2887)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.8652 (0.0035)	0.8652 (0.0035)	0.8652 (0.0035)	0.8652 (0.0035)
macro avg	0.7387 (0.0190)	0.6648 (0.0168)	0.6772 (0.0162)	2849.0000 (0.0000)
weighted avg	0.8701 (0.0079)	0.8652 (0.0035)	0.8572 (0.0052)	2849.0000 (0.0000)

The classification report for the best baseline model shows that the model struggles with classes having few samples, yielding very low precision, recall, and F1-scores. For example, "Youssef Bey Al-Helou," "Father Elias," "Magistrate of Batroun," and "Elias Abdallah" suffer from limited samples and sparse training representation. This data scarcity likely prevents the model from learning meaningful writer-specific patterns, resulting in poor generalization and misidentification during evaluation.

TABLE IV: Classification Report for ResNet50 + Frozen + Attention

Class Name	Value (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9393 (0.0266)	0.9538 (0.0533)	0.9462 (0.0363)	65.0000 (0.0000)
Angele Ellis	0.7075 (0.0417)	1.0000 (0.0000)	0.8282 (0.0286)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9114 (0.0050)	0.9114 (0.0050)	0.9114 (0.0050)	0.9114 (0.0050)
macro avg	0.8146 (0.0225)	0.7792 (0.0171)	0.7795 (0.0169)	2849.0000 (0.0000)
weighted avg	0.9135 (0.0066)	0.9114 (0.0050)	0.9069 (0.0043)	2849.0000 (0.0000)

TABLE V: Classification Report for DenseNet201 Fine-tuned + Last 25 Layers + No Attention

Class Name	Value (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9800 (0.0228)	0.9846 (0.0154)	0.9821 (0.0115)	65.0000 (0.0000)
Angele Ellis	0.8654 (0.0449)	0.8889 (0.0481)	0.8767 (0.0418)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.6667 (0.5774)	0.3333 (0.2887)	0.4444 (0.3849)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	1.0000 (0.0000)	0.4444 (0.1925)	0.6000 (0.1732)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.6667 (0.5774)	0.6667 (0.5774)	0.6667 (0.5774)	2.0000 (0.0000)
Father Elias	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	1.0000 (0.0000)
Magistrate of Batroun	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9404 (0.0093)	0.9404 (0.0093)	0.9404 (0.0093)	0.9404 (0.0093)
macro avg	0.8659 (0.0165)	0.8362 (0.0293)	0.8353 (0.0230)	2849.0000 (0.0000)
weighted avg	0.9430 (0.0085)	0.9404 (0.0093)	0.9375 (0.0096)	2849.0000 (0.0000)

TABLE VI: Classification Report for DenseNet201 Fine-tuned + Last 25 Layers + Attention

Class Name	Value (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9750 (0.0228)	0.9897 (0.0089)	0.9823 (0.0158)	65.0000 (0.0000)
Angele Ellis	0.7951 (0.0776)	0.9167 (0.0833)	0.8470 (0.0287)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.5000 (0.5000)	0.3333 (0.2887)	0.3889 (0.3469)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	0.8000 (0.3464)	0.6667 (0.0000)	0.7000 (0.1732)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.6667 (0.3333)	0.6667 (0.2887)	0.6222 (0.2037)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9479 (0.0112)	0.9479 (0.0112)	0.9479 (0.0112)	0.9479 (0.0112)
macro avg	0.8897 (0.0237)	0.8703 (0.0266)	0.8659 (0.0272)	2849.0000 (0.0000)
weighted avg	0.9520 (0.0106)	0.9479 (0.0112)	0.9462 (0.0116)	2849.0000 (0.0000)

TABLE VII: Classification Report for DenseNet201 + From Scratch + No Attention

Class Name	Value (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9850 (0.0149)	0.9949 (0.0089)	0.9898 (0.0043)	65.0000 (0.0000)
Angele Ellis	0.8990 (0.0364)	0.9722 (0.0481)	0.9332 (0.0234)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.6667 (0.5774)	0.5000 (0.5000)	0.5556 (0.5092)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	1.0000 (0.0000)	0.6667 (0.3333)	0.7667 (0.2517)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.6667 (0.5774)	0.5000 (0.5000)	0.5556 (0.5092)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	1.0000 (0.0000)	0.6667 (0.2887)	0.7778 (0.1925)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9711 (0.0016)	0.9711 (0.0016)	0.9711 (0.0016)	0.9711 (0.0016)
macro avg	0.9242 (0.0147)	0.9042 (0.0149)	0.9052 (0.0152)	2849.0000 (0.0000)
weighted avg	0.9702 (0.0033)	0.9711 (0.0016)	0.9687 (0.0026)	2849.0000 (0.0000)

TABLE VIII: Classification Report for DenseNet201 + From Scratch + Attention

Class Name	Value (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9949 (0.0087)	0.9897 (0.0089)	0.9923 (0.0001)	65.0000 (0.0000)
Angele Ellis	0.9209 (0.0037)	0.9722 (0.0481)	0.9456 (0.0250)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	1.0000 (0.0000)	0.6667 (0.2887)	0.7778 (0.1925)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	0.9167 (0.1443)	0.7778 (0.1925)	0.8190 (0.0330)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.8889 (0.1925)	0.6667 (0.2887)	0.7111 (0.0770)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	1.0000 (0.0000)	0.6667 (0.2887)	0.7778 (0.1925)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9674 (0.0046)	0.9674 (0.0046)	0.9674 (0.0046)	0.9674 (0.0046)
macro avg	0.9298 (0.0153)	0.9030 (0.0283)	0.9060 (0.0240)	2849.0000 (0.0000)
weighted avg	0.9679 (0.0049)	0.9674 (0.0046)	0.9653 (0.0052)	2849.0000 (0.0000)